Check for updates

ASME Journal of Engineering for Gas Turbines and Power
Online journal at:
https://asmedigitalcollection.asme.org/gasturbinespower

Experimental Identification of the Operability Limits in Exhaust Gas Recirculation Rates Under Transient Control for a Microgas Turbine

Vincent Thielens[1]
Thermal Engineering and Combustion Unit,
University of Mons,
Mons 7000, Belgium;
Mechanics and Thermal Processes,
Engie R&I Laborelec,
Linkebeek 1630, Belgium
e-mail: vincent.thielens@umons.ac.be

Frederiek Demeyer
Mechanics and Thermal Processes,
Engie R&I Laborelec,
Linkebeek 1630, Belgium

Klaus Peter Gleigle
Institute of Combustion Technology,
German Aerospace Center,
Stuttgart 70569, Germany

Peter Kutne
Institute of Combustion Technology,
German Aerospace Center,
Stuttgart 70569, Germany

Ward De Paepe
Thermal Engineering and Combustion Unit,
University of Mons,
Mons 7000, Belgium

Exhaust gas recirculation (EGR) has been identified as a key adaptation for gas turbines to mitigate the efficiency penalty of amine-based postcombustion carbon capture. By reducing the exhaust gas flow rate and increasing CO_2 concentration, EGR enhances capture efficiency while also significantly lowering NO_x emissions and residual O_2, both of which influence amine stability. However, the impact and limitations of EGR in dynamic operation remain largely unexplored, representing a critical gap in operability assessments. Understanding these aspects can facilitate the development of control strategies and optimization algorithms where EGR serves as an additional control variable to mitigate pollutant emission peaks. To address this, a 3 kW_e microgas turbine (mGT) setup with enhanced EGR was tested under sudden EGR variations, while monitoring emissions. Additionally, emissions were measured during hot starts with different pre-applied recirculation levels. Results indicate that NO_x peaks occur during EGR transitions, and high EGR rates prevent successful turbine startup. Identifying these transient behaviors, in terms of emissions and combustion stability, provides initial insights into defining operability boundaries for EGR application. The development of control algorithms to manage EGR levels is left for future work. [DOI: 10.1115/1.4069725]

Keywords: microgas turbine, exhaust gas recirculation, emissions, CO, NO_x, transient operation

Introduction

On the technological pathway to carbon neutrality, exhaust gas recirculation (EGR) increases the amine-based carbon capture efficiency by decreasing the mass flow of exhaust gases and increasing the CO_2 concentration, as asserted by De Paepe et al. [1]. Additionally, EGR also enables to decrease NO_x and O_2 levels, both responsible for amine degradation as presented by Fostås et al. [2]. However, high EGR rates are associated with conditions close to stoichiometry leading to carbon monoxide emission and risk of flameout as experimented by Thielens et al. [3]. Even if Verhaeghe et al. [4] have shown the strong benefit of EGR to carbon capture, the transient factors limiting recirculation have still to be investigated.

The impact of transient operations on gas turbine emissions is not well documented in the literature. Environmental policies, such as those developed by the European Union [5], regulate gas turbine emissions based on daily or monthly averages, even though continuous monitoring is required. As a result, peak emissions receive less attention due to the averaging process, and emissions during transient operations have not been a primary focus. However, growing environmental concerns are likely to increase the monitoring of transient emissions. In the context of carbon capture, peaks in NO_x and CO concentrations are particularly relevant as they can affect amine stability and overall system performance.

Rosfjord and Cohen studied the impact of transient operation on turbine combustors and highlighted the risk of blowout in staged combustors [6]. When staging is used to reduce NO_x emissions by distributing the temperature field, reductions in fuel flow can create excessively lean regions, leading to flame blowout. However, their study focused only on monotonic and oscillatory variations of the air/fuel flow rate, without considering exhaust gas recirculation [6].

In their report, Pavri and Moore [7] stated that while CO emissions can remain below 10 ppmvd during steady-state operation, higher levels are observed during ignition and load increases. However, as the share of renewable in the grid continues to grow, such load variations are expected to become more frequent.

Tenango-Pirín et al. numerically investigated combustion instabilities under varying operating conditions [8]. Their study revealed that pollutant formation, particularly CO and NO_x, is highly sensitive to inlet air pressure. Variations in air mass flow influence flame dynamics, leading to the formation of intermediate

Turbo Expo: Turbomachinery Technical Conference & Exposition (GT2025), June 16–20, 2025. GT2025.
[1]Corresponding author.
Manuscript received July 15, 2025; final manuscript received August 12, 2025; published online October 25, 2025. Editor: Jerzy T. Sawicki.

Copyright © 2026 by ASME

Downloaded from http://asmedigitalcollection.asme.org/gasturbinespower/article-pdf/148/3/031012/7536546/gtp-25-1354.pdf by University of Cincinnati user on 31 December 2025

combustion products. Regarding NO_x formation, their simulations showed that emissions peak when pressure perturbations are strongest. Consequently, changes in operating conditions increase the frequency of pressure fluctuations, thereby enhancing pollutant generation.

Thielens et al. [3] experimentally investigated advanced EGR rates up to the point of flameout. Their study first highlighted a key distinction between emissions and concentration. For NO_x, the concentration follows a parabolic trend, peaking at around 60% EGR before declining. However, due to the reduction in exhaust mass flow enabled by EGR, total NO_x emissions (the product of concentration and exhaust flow rate) continuously decrease as recirculation increases. Regarding carbon monoxide, its concentration rises steadily with increasing EGR, driven by O_2 depletion, which promotes incomplete combustion.

While some studies have examined pollutant evolution during transient operations, their interaction with exhaust gas recirculation has never been experimentally investigated or documented. To address this gap, a modified microgas turbine (mGT) setup with enhanced EGR capabilities has been equipped with an exhaust gas analyzer to assess the impact of hot startup and EGR variations on emissions. First, the steady-state concentration levels of O_2, CO_2, NO_x, and CO are characterized, followed by an analysis of their evolution across a wide range of operating conditions (e.g., hot start and EGR change).

Methodology

Since conducting the experimental campaign on industrial-scale gas turbines is not economically feasible, an mGT was chosen for its adaptability and lower cost. The 3 kW_e EnerTwin from MTT® is equipped with an EGR valve (11–12 in Fig. 1) controlled by the machine's internal regulation system. However, this built-in valve alone is insufficient to achieve high EGR rates. To overcome this limitation, the setup was modified with an external EGR loop (13–14) and a back-pressure valve (15–16). The external EGR loop is made with 100 mm diameter spiral tubes to provide a section high enough to decrease pressure losses through the recirculation loop. Both the internal (11–12), external (13–14), and back-pressure (15–16) valves are of the butterfly type. While butterfly valves do not provide precise closure angle or flow control, they offer a compact and cost-effective solution. When fully open, their disk aligns with the flow, minimizing pressure drops.

In the modified setup described in Fig. 1, incoming air (1) is preheated using an external electric resistor (2–3). The air is then compressed from atmospheric pressure (4) to 2.4 bar (5) and further preheated in the internal recuperator before entering the combustion chamber. Fuel from the gas line (17) is compressed (18) and injected

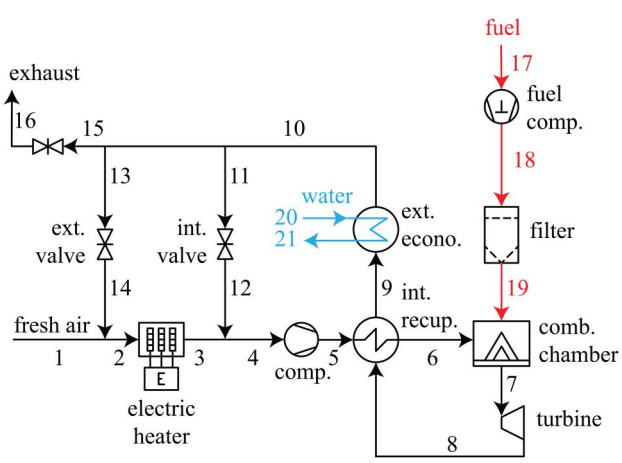

Fig. 1 The 3 kW_e mGT EnerTwin from MTT® has been modified with an electric preheater and an external EGR loop for achieving EGR rates exceeding 80%

Table 1 Accuracy on the different measurement range of the Testo 350® gas analyzer

Element	Accuracy	Range
O_2	±0.2% of read value	—
CO	±10 ppm	<199 ppm
	±5% of read value	200–2000 ppm
	±10% of read value	2001 ppm <
NO	±5 ppm	<99 ppm
	±5% of read value	100–1999.9 ppm
	±10% of read value	2000 ppm <
NO_2	±2 ppm	<39.9 ppm
	±5% of read value	40 ppm <
CO_2	±0.3% ±1% of read value	0–25%
	±0.5% ±1.5% of read value	25% <

into the combustion chamber. After combustion, the exhaust gases expand through the radial turbine (8), and their residual energy is utilized to heat the inlet water (14) to the desired temperature for cogeneration (15). Following the heat exchanger, the cooled exhaust gases can be recirculated either after the electric resistor (11) or before it (13).

The mGT setup is equipped with sensors to measure pressure and temperature at various locations. The pressure transmitter comes from Huba Control® (Walddorfhäslach, Germany) while the temperatures are either measured with Tasseron® (Nootdorp, The Netherlands) NTC sensor or with Omega® (Germany Deckenpfronn) K-Type thermocouple. The dry composition of the exhaust gases (16) is analyzed using a Testo 350® (Ternat, Belgium) gas analyzer, which measures O_2, CO_2, NO, NO_2, and CO with the accuracies specified in Table 1. Before going to the measurement cells, the gas is extracted in the form of a partial sample that is filtered, cooled, and dried. O_2 is measured with an electrochemical sensor with two electrodes. The flue gas reaches the cathode through the gas-permeable membrane where OH^- ions are released. The electrical current produced is proportional to the oxygen concentration, and the voltage across a resistor is measured. CO and NO_x concentrations are also measured with an electrochemical sensor but with three electrodes (working, reference, and counter electrodes) where H^+ is produced. CO_2 is measured with a nondispersive infrared sensor using a broadband lamp source. To decrease the uncertainty on the steady-state measurements, data are sampled and averaged to reduce the variance as explained by Miller [9].

The turbine operates with natural gas during the tests, with an average composition provided by Fluxys, expressed as $C_{1.063}H_{4.045}N_{0.020}O_{0.030}$. Under stoichiometric conditions, the maximum achievable dry molar concentration of CO_2 in the exhaust gases is 12.1%, making higher concentrations theoretically unattainable. Given that only an average composition of natural gas is available, the 95% confidence interval for the maximum achievable dry molar concentration of CO_2 ranges from 11.85% to 12.19%. Since no sensors directly measure the recirculated flow rate, the EGR ratio is determined based on the measured O_2 dry concentration in the exhaust gases, assuming that the mass flow

Table 2 Status from 0 to 7 are used to define the different scenarios of valves opening, leading to different calculated EGR ratios

Status	Internal 11–12	External 13–14	Back-pressure 15–16	EGR
0	0%	0%	100%	1%
1	50%	0%	100%	21%
2	100%	0%	100%	30%
3	100%	100%	100%	64%
4	100%	100%	75%	64%
5	100%	100%	50%	66%
6	100%	100%	25%	72%
7	100%	100%	0%	81%

Downloaded from http://asmedigitalcollection.asme.org/gasturbinespower/article-pdf/148/3/031012/7536546/gtp-25-1354.pdf by University of Cincinnati user on 31 December 2025

Table 3 The EGR transitions experimented between the states described in Table 2 have been represented by a mark "X"

From/To	0	1	2	3	4	5	6	7
0		X	X	X	X	X	X	X
1			X					
2				X				
3					X	X	X	X
4						X	X	X
5							X	X
6								X
7								

through the turbine remains constant. Complete combustion is assumed, as confirmed by the low equivalence ratio and the minimal presence of CO in the exhaust gases.

The test campaign was conducted to analyze emissions across different EGR rates. Following a 2-h preheating phase, various valve configurations were applied, as detailed in Table 2, to achieve the different scenarios:

- the inner valve (11–12) was set to fully closed, fully open, or half-open positions;
- the external valve (13–14) was either fully closed or fully open;
- the back-pressure valve (15–16) was adjusted through four different closure levels, ranging from fully open to near flameout conditions.

The combination of the previous states leads to eight status from which it is switched as represented in Table 3.

Once the mGT components (e.g., the heat exchangers and the turbomachinery) are fully preheated, startup tests are conducted with the eight predefined configurations applied directly. The preheating phase is considered complete when all sensor readings stabilize and show no further variation. Emissions in the exhaust gases (16) are then continuously monitored throughout the process.

Results and Discussion

To assess the impact of EGR on transient operations, the exhaust gas composition is first measured in steady-state for each configuration described in Table 2. These steady-state measurements provide a reference for evaluating the changes induced by transient conditions. Emissions are then monitored during both EGR ratio transitions and hot starts to analyze their dynamic effects.

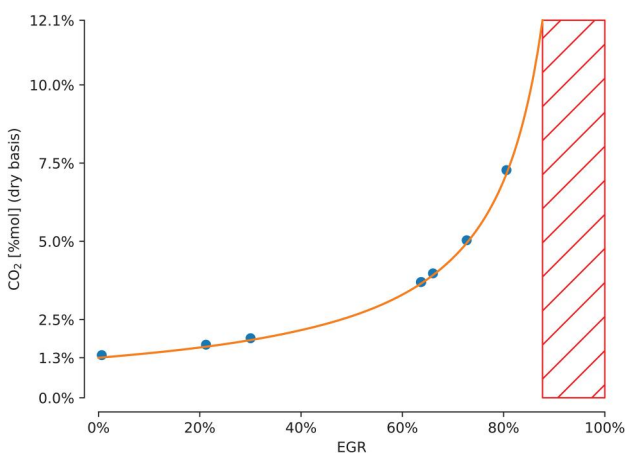

Fig. 2 The CO_2 dry measurements in the exhaust gases (dots) follow the analytical combustion law (line) among the different EGR rates. Starting from 1.3%, the CO_2 dry concentration can theoretically reach 12.1% at stoichiometry. The accuracy of the measurement device is presented in Table 1.

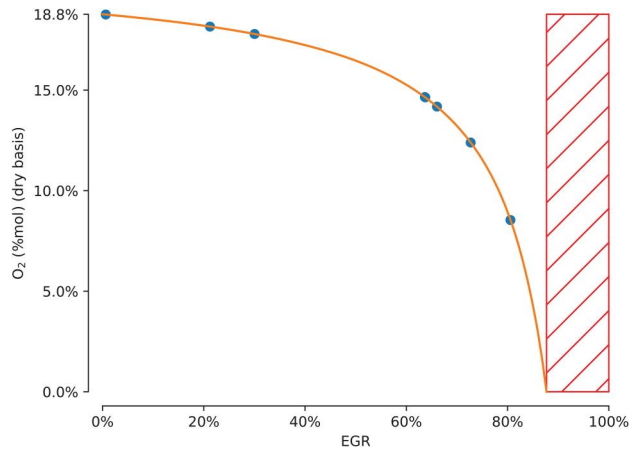

Fig. 3 The O_2 dry measurements in the exhaust gases (dots) follow the analytical combustion law (line) among the different EGR rates. Starting from 18.8%, the O_2 dry concentration decreases to zero at stoichiometry. The accuracy of the measurement device is presented in Table 1.

Steady-State. After the preheating phase, different EGR levels are applied, according to Table 2, and the concentrations of O_2, CO_2, CO, NO, and NO_2 are measured with the gas analyzer. The dry molar concentrations of CO_2 and O_2 follow the theoretical law as represented in Figs. 2 and 3, respectively.

The CO_2 concentration starts at 1.3% with no EGR and reaches a maximum of 7.3% at 81% EGR (Fig. 2). Due to the high excess air used ($\lambda = 8.5$) at 0% EGR, the oxygen content in the exhaust gases remains relatively high, at 18.8%, as shown in Fig. 3. At the highest tested EGR level, the dry molar O_2 concentration decreases to 8.5%, resulting in a normalized air-fuel ratio of 1.6. Under stoichiometric conditions, the CO_2 concentration could theoretically reach 12.1%, with oxygen levels dropping to zero. However, operating at such rich conditions would be too restrictive for the combustor. The combustor architecture divides the chamber into a primary and a secondary zone, with dilution air introduced in the latter. Supplying the inlet with a stoichiometric air–fuel mixture would lead to locally incomplete conditions in the primary zone, given the air fraction allocated to dilution in the secondary zone [10].

The evolution of CO emissions is depicted in Fig. 4. Initially measured at 27 ppm, CO concentrations remain below 30 ppm up to 60% EGR. Beyond this point, CO levels increase sharply due to oxygen depletion in the combustion chamber, as previously reported

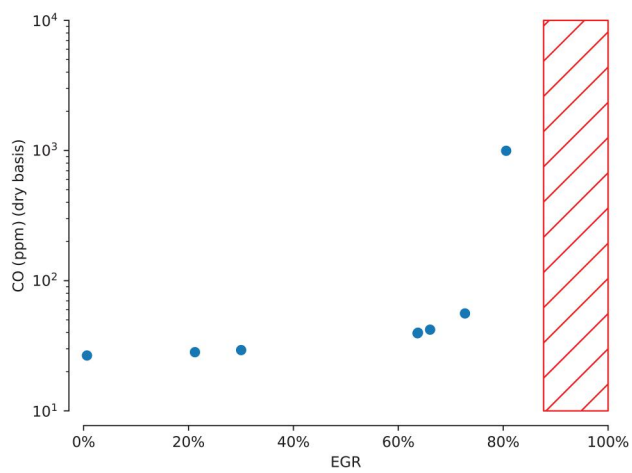

Fig. 4 The CO concentration increases exponentially in the exhaust gases. Starting from a concentration between 20 and 30 ppm, CO reaches 1000 ppm at 80% EGR. The accuracy of the measurement device is presented in Table 1.

Downloaded from http://asmedigitalcollection.asme.org/gasturbinespower/article-pdf/148/3/031012/7536546/gtp-25-1354.pdf by University of Cincinnati user on 31 December 2025

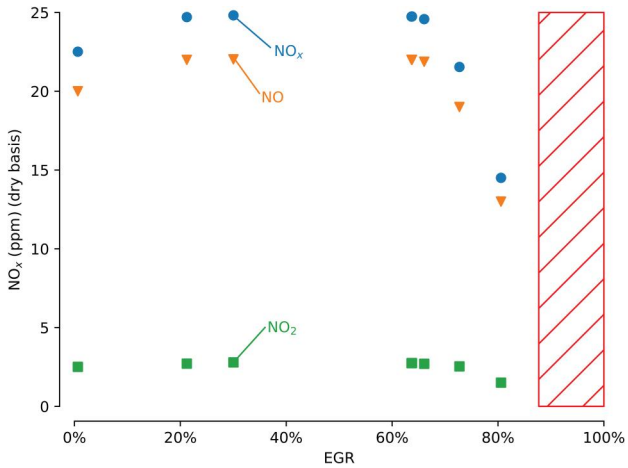

Fig. 5 After an increase, due to the additional residence time implied by EGR, the O_2 depletion drastically decreases the NO_x production. The accuracy of the measurement device is presented in Table 1.

by ElKady et al. [11]. Additionally, CO_2 dissociation, following the mechanisms described by Masri et al. [12], has been identified as a contributing factor to CO formation.

In Fig. 5, the evolution of nitrogen oxides follows a parabolic trend as a function of the recirculation rate. Starting at 22 ppm without EGR, NO_x levels increase to a peak of approximately 25 ppm at 50% EGR. This rise is attributed to the increased effective residence time caused by exhaust gas recirculation. Since a fraction of the exhaust gases is recirculated, it passes through the combustion chamber multiple times, effectively extending its exposure to high-temperature conditions. The average number of recirculations, n_r, is determined by considering the balance between the fresh air fraction $(1 - \mathrm{EGR})$ and the recirculated fraction (EGR). At each iteration i, the flue gases have undergone n_i recirculations, as defined in Eq. (1). This sequence converges to the final recirculation number, n_r, as shown in Eq. (2). Beyond the peak NO_x emissions, oxygen depletion becomes the dominant factor, significantly reducing NO_x formation. As seen in Fig. 5, NO_2 concentrations remain lower than NO levels, as the oxidation kinetics do not allow complete conversion from NO to NO_2 under the given conditions

$$n_i = (1 - \mathrm{EGR}) + (x_{i-1} + 1)\mathrm{EGR} = \sum_{j=0}^{i} \mathrm{EGR}^j \tag{1}$$

$$n_r = \lim_{i \to \infty} n_i = \frac{1}{1 - \mathrm{EGR}} \tag{2}$$

Exhaust Gas Recirculation Transient. During steady-state operation, the exhaust gas recirculation rate is suddenly adjusted between different levels. The changes in EGR rates are detailed in Table 3, and the corresponding O_2, CO_2, NO_x, and CO concentrations in the exhaust gases (16 in Fig. 1) are presented. Both CO_2 and O_2 concentrations do not exhibit peak emissions when the EGR rate is adjusted. Instead, their values increase gradually from one steady-state to the next. However, the time required for stabilization increases as the target EGR level becomes higher. The settling times for these transitions are presented in Table 4, showing the time it takes for the exhaust gas concentrations to stabilize. The settling time, $t_{0.2}$, is defined as the duration from when the dry O_2 concentration begins to change until it stabilizes within a 0.2% absolute window around the final steady-state concentration. As shown in Table 4, the settling time $t_{0.2}$ can reach up to 62 s for the largest transitions and decreases as the change in EGR rate becomes smaller. This increase in stabilization time is linked to the time required for the recirculation process to fully establish itself.

Table 4 The settling time $t_{0.2}$ in seconds around 0.2% abs. of the final dry O_2 composition decreases as the transition step gets closer

From/To	0	1	2	3	4	5	6	7
0		6	7	28	28	28	28	62
1			4	-	-	-	-	-
2				21	-	-	-	-
3					-	5	22	59
4						5	22	59
5							21	55
6								53
7								

The transition states refer to the nomenclature established in Table 2.

Changes in the EGR ratios lead to distinct NO_x emissions during the transient phase, categorized into one of the three types presented in Fig. 6. Starting from zero recirculation, various EGR levels were applied, and the corresponding NO_x emissions are summarized in Table 5. For transitions to EGR rates below 30%, the NO_x emissions evolve smoothly from the previous steady-state level to the next, following the behavior depicted by type 1 in Fig. 6. However, experiments revealed a peak in NO_x emissions when transitioning from 0% to EGR levels above 60%. This peak increases as the EGR rate is raised. For EGR levels exceeding 70%, resulting in a NO_x concentration lower than that observed without EGR at steady-state (as shown in Fig. 5), the emission evolution follows type 3, as defined in Fig. 6. For intermediate levels, before reaching 70% EGR, the emission evolution corresponds to type 2.

In contrast to the previous results, where transitions occurred from a state with no EGR applied, increasing the EGR level from an already applied recirculation results in different behavior. When transitioning from state 3 (64% EGR) to states 5 (66% EGR), 6 (72%

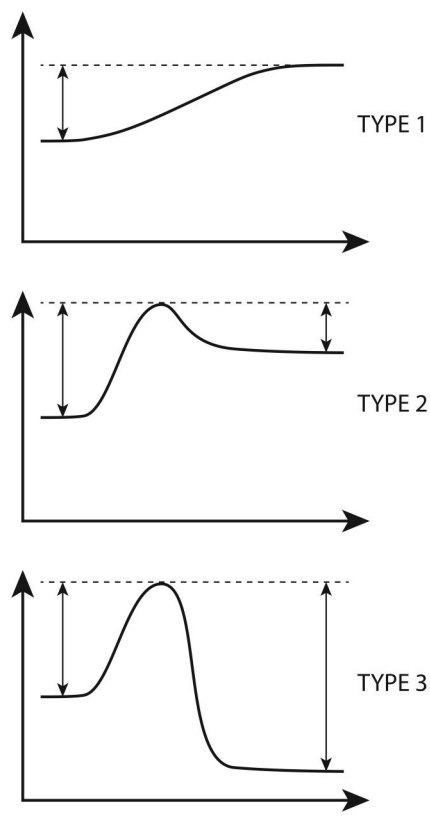

Fig. 6 The NO_x transient evolution can be categorized in three different types, depending on the change in steady-state value before and after the peak

Downloaded from http://asmedigitalcollection.asme.org/gasturbinespower/article-pdf/148/3/031012/7536546/gtp-25-1354.pdf by University of Cincinnati user on 31 December 2025

Table 5 The changes from state 0 to higher EGR ratios lead to NO_x peaks

From	To	To EGR	$NO_{x,\,inc.}$ (ppm)	$NO_{x,\,peak}$ (ppm)	$NO_{x,\,dec.}$ (ppm)	Type
0	1	21%	2	24.8	0	1
0	2	30%	2	24.9	0	1
0	3	64%	6	28.2	3	2
0	4	64%	6	28.2	3	2
0	5	66%	6	28.3	4	2
0	6	72%	7	29.4	8	3
0	7	81%	8	30.6	16	3

Those peaks can be classified in three categories presented in Fig. 6 depending on the shape of the transient curve.

Table 6 Starting from an advanced EGR rate (64%) allows a lower NO_x peak before the NO_x depletion

From	To	To EGR	$NO_{x,\,inc.}$ (ppm)	$NO_{x,\,peak}$ (ppm)	$NO_{x,\,dec.}$ (ppm)	Type
3	5	66%	0	23.9	0	1
3	6	72%	1	24.9	3	3
3	7	81%	1.5	25.2	11	3

EGR), or 7 (81% EGR), the NO_x emission peak decreases. As shown in Table 6, the peak emissions required to reach the same EGR level from state 3 are, on average, 5 ppm lower than those encountered when starting from state 0.

Looking at carbon monoxide, the emissions evolve from the previous level to the one imposed by the new EGR level according to type 1 (Fig. 6). Risk of CO peak emission is thus not a constraint limiting the EGR change.

No significant change in electric power production is observed during the transition between EGR levels. Thielens et al. [3] previously demonstrated that increasing EGR can slightly enhance power production due to the cooling effect of exhaust gases when mixed with ambient air. However, in this setup, the preheating system ensures a constant temperature at the compressor inlet, eliminating this effect. Additionally, since no significant pressure drop occurs in the recirculation loop, the power output of the microgas turbine remains unchanged throughout the transition between EGR levels.

Hot Start. A hot start is defined as the startup of the mGT immediately after a shutdown, meaning that all components (e.g., heat exchangers) remain preheated. Compared to a cold start, hot start testing is more convenient, as no cooling time is required between experiments at different EGR levels. Additionally, hot starts provide a lower bound for CO emissions since the preheated combustion chamber enhances combustion efficiency, reducing the amount of incomplete products.

During the startup phase, the shaft is driven electrically to pressurize the combustion chamber. Once the required conditions are met, fuel is injected and ignited, causing the oxidizer concentration to drop from the ambient 21%, as illustrated in Fig. 7. Following the acceleration phase, the mGT stabilizes at its nominal operating conditions. As the gas flow gradually decreases, the oxygen concentration in the exhaust gases rises until it reaches the steady-state value, as shown in Fig. 3.

Regarding CO_2, its evolution follows an opposite trend to the one of oxygen, as shown in Fig. 8. Before combustion begins, no carbon-containing species are present in the exhaust gases. However, during the acceleration phase, the additional fuel consumption leads to a sharp increase in CO_2 concentration, resulting in a peak significantly higher than the steady-state value.

For both oxygen and carbon dioxide, the stabilization time of their concentrations in the exhaust gases increases as a larger fraction of exhaust gases is recirculated from the startup phase. Additionally, the two-zone architecture of the combustion chamber prevents ignition at extreme EGR rates (e.g., 81%). In the highest EGR case referenced in Table 2, a flameout was observed during startup due to insufficient oxygen availability in the primary combustion zone.

The evolution of CO emissions is depicted and justified in Fig. 9. Before ignition, CO is absent from the exhaust gases as no combustion occurs. Upon ignition, a sharp peak in CO emissions appears, followed by a rapid decline as fresh air enters the combustion chamber and increased heat release enhances combustion completeness. This initial CO peak is attributed to the low equivalence ratio at ignition. As the turbine shaft accelerates, more air flows into the mGT, reducing CO formation. Once sufficient air is available to sustain complete combustion, flame dilution and the resulting temperature decrease become the dominant factors influencing CO production. Consequently, CO levels evolve with the air–fuel ratio, as greater combustion dilution lowers flame temperature, promoting incomplete combustion and increased CO formation.

The impact of EGR on CO emissions is illustrated in Fig. 10, where the initial peak is clearly represented. EGR increases CO emissions during the starting phase. Although no CO should be present before ignition, some was detected at the highest EGR level due to noise in the gas analyzer. Since CO is toxic to the analyzer measurement cells, a dilution factor of five is applied to protect them, which increases baseline noise. Additionally, if residual CO

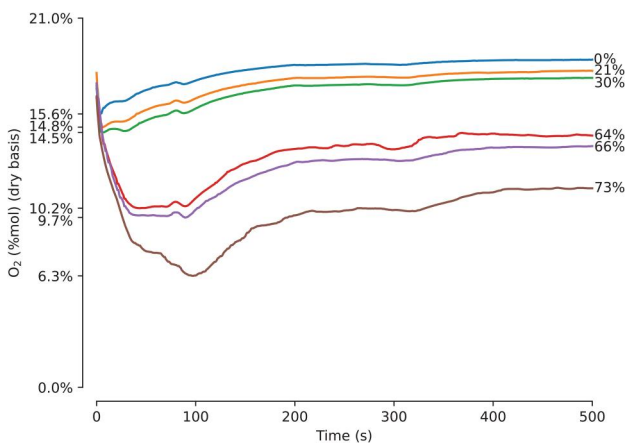

Fig. 7 During the start phase, an oxygen depletion occurs before increasing to reach the steady-state value. The depletion is deeper as the EGR rate gets higher. The accuracy of the measurement device is presented in Table 1.

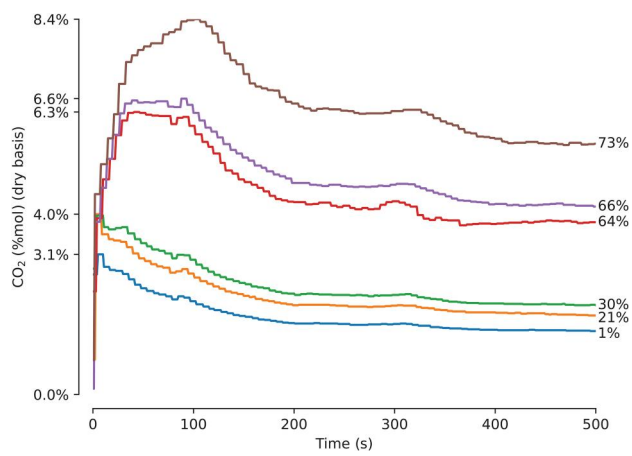

Fig. 8 During the start phase, a CO_2 peak is observed before the concentration decreases to reach the steady-state value. The accuracy of the measurement device is presented in Table 1.

Downloaded from http://asmedigitalcollection.asme.org/gasturbinespower/article-pdf/148/3/031012/7536546/gtp-25-1354.pdf by University of Cincinnati user on 31 December 2025

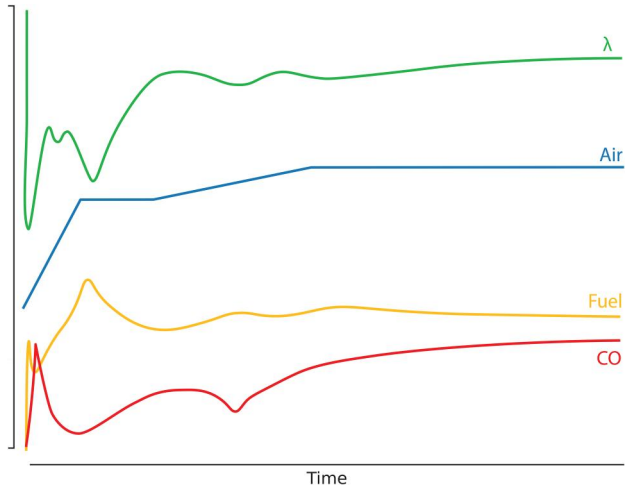

Fig. 9 After the CO peak produced by the ignition, the highly diluted combustion slightly cools the flame, thus increasing so the CO produced

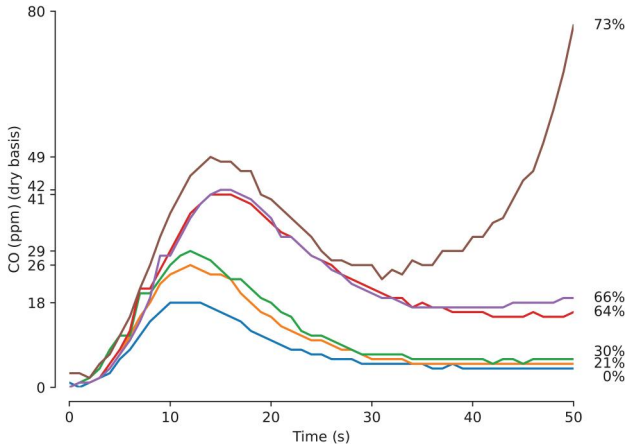

Fig. 10 The first CO peak encountered at startup increases with the EGR rate applied. The accuracy of the measurement device is presented in Table 1.

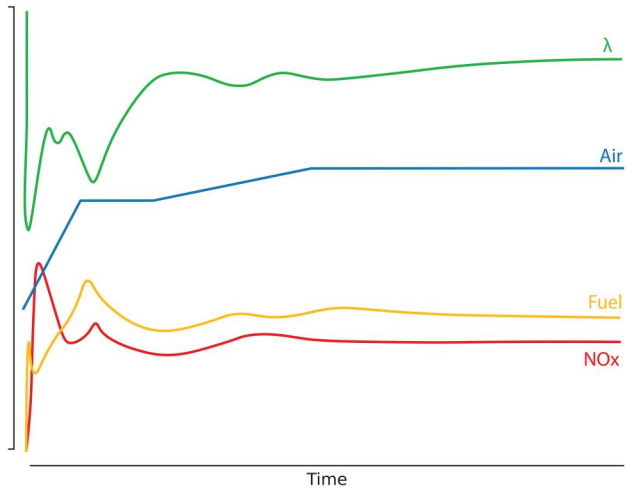

Fig. 11 The NO_x emission faces a peak at the ignition phase due to the nonoptimal condition in the combustion chamber. The NO_x level then decreases and stabilizes.

remains from the previous operating period, a high EGR rate traps a fraction of these pollutants within the recirculation loop. As shown in Fig. 10, the CO peak height increases with higher recirculation rates.

The evolution of NO_x concentration in the mGT during the startup phase is shown in Fig. 11. A peak in NO_x emissions occurs at ignition, primarily due to the extended residence time caused by the low air flow rate and suboptimal combustion conditions. Limited air flow results in poor mixing and the formation of hot spots, both of which contribute to increased NO_x production. Following this initial peak, the additional air dilutes the combustion process and lowers the flame temperature. Since NO_x formation is primarily governed by the Zeldovich's mechanism, the temperature reduction directly decreases NO_x emissions. Consequently, NO_x levels evolve inversely with the equivalence ratio.

Although exhaust gas recirculation slightly reduces the initial NO_x peak during the starting phase (Fig. 12), the effect remains minimal. This is because, at ignition, the combustion process has not yet fully experienced the impact of recirculated exhaust gases, such as the reduction in oxygen concentration and the increased presence of CO_2 in the combustion chamber. While considering the maximum EGR of 73%, the reduction in NO_x becomes visible due to reduced O_2 contents.

Good Practices on Control Strategies. Environmental policies are expected to become increasingly stringent in the coming years, regulating not only the daily average emissions but also peak emissions. Moreover, the growing share of renewable energy in the grid will force gas turbines to operate more frequently as peak units, significantly increasing transient operations. Additionally, NO_x and CO have been identified as pollutants that impact the chemical stability of amines. While their exact influence has yet to be quantified, experimental work has highlighted best practices and key considerations for managing EGR effectively.

- *Maximizing EGR under the CO emission limit in steady-state operation*: Achieving the highest feasible level of exhaust gas recirculation during steady-state operation helps reduce the overall exhaust gas mass flow, increase CO_2 concentration, and lower O_2 and NO_x concentrations. These effects benefit amine-based carbon capture but are constrained by CO emission limits.
- *Starting without EGR*: Although initiating gas turbine operation with EGR does not impact NO_x emissions, it increases the peak in CO emissions. To avoid unnecessary CO peaks, it is preferable to start without EGR and gradually increase the recirculation rate after the initial CO peak has been encountered.

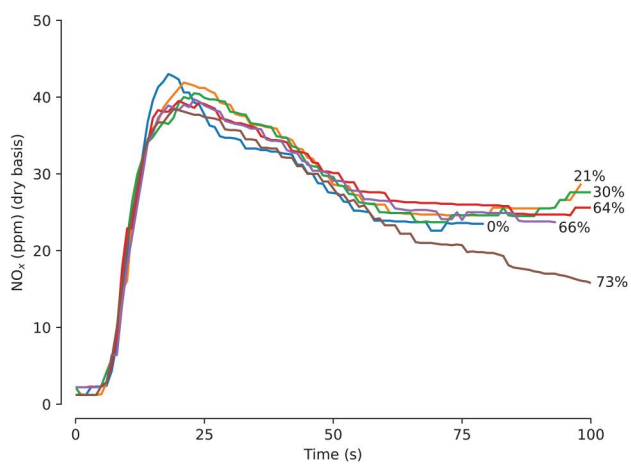

Fig. 12 Starting with EGR does not bring significant reduction on the first NO_x peak. The accuracy of the measurement device is presented in Table 1.

Downloaded from http://asmedigitalcollection.asme.org/gasturbinespower/article-pdf/148/3/031012/7536546/gtp-25-1354.pdf by University of Cincinnati user on 31 December 2025

- *Gradually adjusting the EGR rate*: While CO emissions are not a limiting factor during EGR transitions, a NO_x peak can occur. Therefore, it is more beneficial to adjust the EGR level progressively rather than making abrupt changes.

Conclusion

Exhaust gas recirculation is employed in amine-based carbon capture to increase CO_2 concentration and reduce the exhaust gas mass flow rate. However, EGR also influences NO_x and CO emissions, both in steady-state and transient conditions. To address the lack of experimental data on EGR's impact on emissions during transient phenomena, a 3 kWe mGT from MTT® was modified and equipped with an external EGR loop. Seven operating states, each characterized by a different EGR rate, were defined. Emissions were measured using a gas analyzer during transitions as well as during startup.

In steady-state operation, the dry molar concentrations of O_2 and CO_2 in the exhaust gases align with theoretical predictions. However, EGR significantly affects NO_x and CO production. Regarding NO_x, increasing the EGR rate initially leads to a rise in NO_x concentration before a significant reduction due to oxygen depletion. This depletion also causes a sharp increase in CO levels at high EGR rates.

During transient operation, NO_x peaks occur when transitioning between EGR levels, with peak intensity increasing as the difference between EGR levels grows. In contrast, CO emissions transition more gradually between steady-state values corresponding to different EGR levels.

While no CO_2 or O_2 peaks are observed during EGR transitions, startup conditions lead to peak emissions. Both CO and NO_x exhibit peaks during startup, but while the NO_x peak remains unaffected by the initial EGR level, the CO peak intensifies with higher EGR levels.

These experiments highlight challenges related to CO and NO_x emissions during startup and phase transitions when applying EGR. Depending on environmental policies and the tolerance of the carbon capture unit, EGR implementation may need to be constrained to minimize or mitigate transient pollutant emissions.

Future Work

While EGR has demonstrated limitations in terms of emissions during transient phases, this study has only focused on the increase in EGR and the startup phase. However, transient phenomena also occur during shutdown and EGR reduction, and they remain unexplored.

With respect to performance, this study has been limited to emissions, and the impact of transients on combined heat and power production and efficiency has not been investigated. Nevertheless, the impact on the heat production is expected to be limited due to the thermal inertia of the system during transients.

Furthermore, given the growing interest in hydrogen, future studies should explore how transitioning from natural gas to hydrogen-containing blends affects cycle emissions. Finally, the research could focus on how O_2 or H_2 enriched combustion could be beneficial toward a further extension of the EGR ratio and as such an enhanced CO_2 concentration in the flue gases. A further enhancement of the CO_2 concentration in flue gases will facilitate amine-based post carbon capture systems worldwide.

Acknowledgment

Vincent Thielens is a Research Fellow of the Fonds de la Recherche Scientifique—FNRS.

Funding Data

- Fonds de la Recherche Scientifique—FNRS (Grant No. J.0033.23 ACCURATE; Funder ID: 10.13039/501100002661).

Data Availability Statement

The datasets generated and supporting the findings of this article are obtainable from the corresponding author upon reasonable request.

Nomenclature

dec = decrease after the peak value
e = electric
inc = increase up to the peak value
peak = peak value
th = thermal

Symbols

n_r = average number of recirculations
$t_{0.2}$ = settling time in seconds to reach a value around 0.2% abs of the steady-state one
λ = dimensionless air–fuel ratio

Abbreviations

EGR = exhaust gas recirculation
GT = gas turbine
mGT = microgas turbine
ppmvd = parts per million volumetric dry

References

[1] De Paepe, W., Montero Carrero, M., Giorgetti, S., Parente, A., Bram, S., and Contino, F., 2016, "Exhaust Gas Recirculation on Humidified Flexible Micro Gas Turbines for Carbon Capture Applications," ASME Paper No. GT2016-57265.
[2] Fostås, B., Gangstad, A., Nenseter, B., Pedersen, S., Sjøvoll, M., and Sørensen, A. L., 2011, "Effects of NO_x in the Flue Gas Degradation of MEA," Energy Procedia, **4**, pp. 1566–1573.
[3] Thielens, V., Demeyer, F., Geigle, K. P., Kutne, P., and De Paepe, W., 2025, "Experimental Investigation of the Emissions and Performance of a Micro Gas Turbine Setup With Enhanced EGR," Appl. Therm. Eng., **267**, p. 125673.
[4] Verhaeghe, A., Mendoza Morales, M. J., Blondeau, J., Demeyer, F., Bricteux, L., and De Paepe, W., 2023, "Thermodynamic Assessment of a Combined Cycle Gas Turbine With Exhaust Gas Recirculation Under Part-Load Operation Toward Carbon Capture Penalty Reduction," ASME Paper No. GT2023-101902.
[5] European Parliament and Council, 2010, "Directive 2010/75/EU on Industrial Emissions (Integrated Pollution Prevention and Control), November," Official Journal of the European Union, Brussels, Belgium.
[6] Rosfjord, T. J., and Cohen, J. M., 1995, "Evaluation of the Transient Operation of Advanced Gas Turbine Combustors," J. Propul. Power, **11**(3), pp. 497–504.
[7] Pavri, R., and Moore, G. D., 2001, "Gas Turbine Emissions and Control," Atlanta: GE Energy Serv., **1**, pp. 1–20.
[8] Tenango-Pirín, O., Espinosa, S., García, J. C., and Rodríguez, J. A., 2024, "Analysis of Gas Turbine Combustor Exhaust Emissions: Effects of Transient Inlet Air Pressure," Energy Technol., **12**(3), p. 2301093.
[9] Miller, S., 2012, *Probability and Random Processes: With Applications to Signal Processing and Communications*, Academic Press, Waltham, MA.
[10] Cirigliano, D., Grimm, F., Kutne, P., and Aigner, M., 2022, "Oxidation-Induced Damage Modeling in Micro Gas-Turbine Combustion Chambers," Procedia Struct. Integr., **42**, pp. 1728–1735.
[11] ElKady, A. M., Evulet, A., Brand, A., Ursin, T. P., and Lynghjem, A., 2009, "Application of Exhaust Gas Recirculation in a DLN F-Class Combustion System for Postcombustion Carbon Capture," ASME J. Eng. Gas Turbines Power, **131**(3), p. 034505.
[12] Masri, A., Dibble, R., and Barlow, R., 1992, "Chemical Kinetic Effects in Nonpremixed Flames of H_2/CO_2 Fuel," Combust. Flame, **91**(3–4), pp. 285–309.

Downloaded from http://asmedigitalcollection.asme.org/gasturbinespower/article-pdf/148/3/031012/7536546/gtp-25-1354.pdf by University of Cincinnati user on 31 December 2025